Student Guide — middle / module-10-herbs-aushadhi

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — Herbs — Auṣadhi (Middle)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *auṣadhi* — ______________________________
- *tulasī* — ______________________________
- *nimba* — ______________________________
- *haridrā* — ______________________________
- *ārdraka* — ______________________________
- *yavānī* — ______________________________
- *pudīnā* — ______________________________

Day-by-day notes

Day 1 — What is Auṣadhi?

What I learned:

One question I have:

Day 2 — Tulsī — Holy Basil

What I learned:

One question I have:

Day 3 — Neem — Bitter Healer

What I learned:

One question I have:

Day 4 — Turmeric — Golden Root

What I learned:

One question I have:

Day 5 — Ginger — Warming Root

What I learned:

One question I have:

Day 6 — Ajwain — Carom Seed

What I learned:

One question I have:

Day 7 — Mint — Cooling Leaf

What I learned:

One question I have:

Day 8 — Herbs and the Doṣas

What I learned:

One question I have:

Day 9 — Build the Class Deck

What I learned:

One question I have:

Day 10 — Final Share and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — What is an *Auṣadhi*?

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can define *auṣadhi*, name at least four of the six herbs the module will cover, and articulate the core safety rule: **food amounts are safe; therapeutic doses are a clinical question**.

Materials

- Six labelled samples laid out at the front of the class: tulsi sprig, neem leaf (dried, no chewing), turmeric root or powder, ginger root, ajwain seeds in a small jar, mint sprig
- Whiteboard
- Six index cards, one per herb, with the Sanskrit + English + botanical name pre-printed

Vocabulary introduced today

- *auṣadhi* (IAST: auṣadhi; lit. "herb, plant used as medicine") — the general term, in Vedic and Ayurvedic Sanskrit, for any plant valued for healing.
- *tulasī, nimba, haridrā, adraka, yavānī, pudīnā* — the six herbs in this module.

Lesson flow

Warm-up (5 min)

Hold up the tulsi sprig. *"How many of you have seen this plant at home?"* Hands.

"How many of you have eaten what's in this jar?" (Pull out ajwain.) Hands.

Establish the premise: most of these are already in their kitchens. We're going to study what's actually in them — and how Indian tradition organised that knowledge.

Core (20 min)

1. **What is an *auṣadhi*?** Write the word on the board.

    - *Auṣadhi* means "plant used for healing." It appears in the Ṛgveda, in a hymn called the *Auṣadhi-sūkta* (Ṛgveda 10.97), where the herbs are addressed directly — as though they were beings with names and personalities. We'll come back to this hymn on Day 8.
    - In *Caraka Saṁhitā, Sūtrasthāna* 1 — one of the foundational texts of Āyurveda — herbs are sorted by their qualities (*guṇa*), their tastes (*rasa*), their heating/cooling potency (*vīrya*), and their effect after digestion (*vipāka*). Don't worry about all four right now — we'll meet *vīrya* on Day 8.

2. **The six herbs we'll study.** Walk along the front bench, holding up each sample:

Sanskrit	English	Botanical	Where you've seen it
tulasī	tulsi, holy basil	*Ocimum tenuiflorum*	garden pot, prayer area
nimba	neem	*Azadirachta indica*	street tree, soap
haridrā	turmeric, haldi	*Curcuma longa*	daal, milk
adraka	ginger, adrak	*Zingiber officinale*	tea, sabzi
yavānī	ajwain, carom	*Trachyspermum ammi*	parathas, achaar
pudīnā	mint, pudina	*Mentha* species	chutney, raita

   Ask the class: *"Which two of these are you most familiar with? Which are you not?"* Show of hands per herb.

3. **The big safety rule** (write on board in capitals — leave it there for the rest of the module):

   FOOD AMOUNTS: SAFE. THERAPEUTIC DOSES: ASK A DOCTOR.

   Explain in one minute: every plant on this table contains chemicals that affect the human body. In small amounts — the pinch you put in food — these chemicals are mostly safe and often helpful. In concentrated form — powders, capsules, oils — they become medicine, and medicine needs supervision.

   "Anyone here ever heard a relative say 'neem is good for you, eat the leaves'? Does anyone want to chew a neem leaf right now?" (They will. Don't let them.) *"Neem in concentrated form can damage the liver — and pure neem leaf is so bitter you'd spit it out anyway. So we'll smell it. We won't eat it."*

Activity (15 min) — Six-herb walkaround

Set up six "stations" — one per herb. Pairs of students rotate through, 1.5 minutes per station. At each:

- **Look.** What shape is the leaf or root?
- **Smell.** Crush a tiny piece (where safe — not neem) and inhale gently from a distance. Don't put it under anyone else's nose.
- **Record** one identifying feature on their workbook page.

At the neem station, leaves are dried and in a sealed bag. **Smell from outside the bag.** No touching the inside.

Last 3 min: pairs report one feature back to the group. Build a quick "first impression" word-cloud on the board.

Wrap-up (5 min)

- Restate the safety rule chorally — the whole class.
- Exit ticket on a slip: write the **Sanskrit name and one identifying feature** for any two herbs you remember.
- Homework: tonight, find as many of the six herbs as you can in your kitchen or balcony. List them.

Student-facing content

An ***auṣadhi*** is a plant used for healing.

We're going to study six of them — every one is already in most Indian kitchens or gardens:

tulasī, *nimba*, *haridrā*, *adraka*, *yavānī*, *pudīnā*.

The rule: food amounts are safe. Therapeutic doses (capsules, powders, oils) are for trained practitioners, not for us.

The early Indian tradition addressed these plants directly in a hymn called the *Auṣadhi-sūkta* (Ṛgveda 10.97). Later, the *Caraka Saṁhitā* (one of the foundational Ayurvedic texts) sorted them by their tastes, qualities, and heating/cooling effect. We'll meet both these ideas over the next two weeks.

Homework

- List which of the six herbs are in your kitchen tonight. Bonus: bring a leaf or a piece (with a parent's permission) to class tomorrow for the tulsi day.

Differentiation

- **One tier up:** Find a *seventh* herb in your kitchen we haven't named (cilantro, fennel, fenugreek, curry leaf, etc.). What might its Sanskrit name be? (Hint: look up *dhānyaka, methi, kariveppilai*.)
- **One tier down:** Just remember three herbs by Friday — tulsi, turmeric, ginger. The other three can come over the next week.

Teacher notes

- Have the safety rule visible on the board for **every class** in this module, not just today.
- A few students will laugh at "addressed the herbs as if they were beings." Use this. It's a glimpse of a world where you talked to plants. We don't have to take the metaphysics literally to take the *attention* seriously.
- The class word-cloud from the walkaround is your fastest read on who already knows what. Use it to plan Days 2–7.

Citation(s) used in this lesson

- *Ṛgveda 10.97* — the *Auṣadhi-sūkta*. Referenced by sūkta number. Real public-domain Vedic source.

- *Caraka Saṁhitā, Sūtrasthāna* 1. Referenced at chapter level. Real foundational Ayurvedic source.

Day 2 — *Tulasī* (Holy Basil)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *tulasī* (*Ocimum tenuiflorum*) by leaf shape and crushed-leaf smell, name eugenol as its principal aromatic compound, and describe two traditional culinary or domestic uses without making therapeutic claims.

Materials

- A live tulsi plant in a pot (ideal); a fresh sprig if a plant isn't available
- A second Lamiaceae plant for comparison if you can — sweet basil (*Ocimum basilicum*) is the easy comparison
- A clove (the spice — also rich in eugenol) for the smell-bridge
- Whiteboard
- Student notebooks

Vocabulary introduced today

- *tulasī* (IAST: tulasī; lit. "the incomparable one") — *Ocimum tenuiflorum*, the herb commonly called holy basil.
- *eugenol* — a phenolic aromatic compound. The strong, slightly clove-like smell of tulsi is largely from eugenol. Also the main aromatic in actual clove (*Syzygium aromaticum*).

Lesson flow

Warm-up (5 min)

Pass the tulsi plant slowly along one row. Each student takes a single leaf between their fingers, rubs it gently, and sniffs.

"What does it smell like? Three words." Take 5–6 responses. Likely answers: "sharp," "minty," "spicy," "clove-y," "medicine-ish."

Now pass the clove. *"Familiar?"* The connection is the molecule. Don't tell them yet — let them notice.

Core (20 min)

1. **Identify it.** Tulsi has:
    - **Leaf shape:** ovate (oval-pointed), about 2–5 cm long, with **toothed margins** (small jagged edges, called "serrate" in botany).

- **Stem:** square in cross-section — characteristic of the **Lamiaceae** family (mint family). All six plants in this family share this. Press a stem between thumb and forefinger and feel the corners.
- **Smell when crushed:** the sharp, clove-like aroma.
- **Common varieties in India:** *Rāma tulasī* (green), *Kṛṣṇa tulasī* (purple-leaved). Same species, different cultivars.

2. **What's inside.** Modern phytochemistry has identified the main aromatic in tulsi as **eugenol** — a phenolic compound ($C_{10}H_{12}O_2$). The same compound dominates clove (the spice). That's why the two smell related.

   Why does the plant make eugenol? Because it's the plant's chemical defence — eugenol discourages many insects from eating the leaves, and it has antibacterial effects in petri-dish studies. The plant doesn't make it for us; we just happened to find it useful.

3. **Traditional uses.** Mention only:

   - As a *kitchen herb* — in chutneys, drinks (tulsi tea — leaves steeped in water), sometimes a few leaves added to dal.
   - As a *household plant* — many Indian homes keep a tulsi plant for its scent and its insect-repelling qualities.
   - As an *aromatic* — in some traditional Ayurvedic preparations for cough and cold support. (Here, drop the safety reminder again — these preparations are formulated by practitioners; we are not recommending self-medication.)

4. **What we won't claim.** Tulsi is sometimes marketed in capsules as an "adaptogen" — a label that means "helps your body handle stress." There is *some* mechanistic evidence (anti-inflammatory effects from eugenol and other compounds), but clinical trial evidence is **mixed**. Saying "tulsi reduces stress" as a blanket claim isn't honest; saying "people have used tulsi as a calming household scent for a long time and there is preliminary evidence of physiological effects" is more accurate.

Activity (15 min) — Smell-and-sketch

- Each student gets a piece of paper and crayons.
- Draw the tulsi leaf — accurate to shape, edge, and size. Label: leaf shape (ovate), edge (serrate), family (Lamiaceae, square stem).
- Below the drawing, write one sentence: *"What I smelled when I crushed the leaf was ____."*
- Below that, the chemistry box: **Active compound: eugenol.**

Optional: pass around the sweet basil (or any other Lamiaceae) for comparison. *"Different leaf? Different smell? They are the same plant family but different species."*

Wrap-up (5 min)

- Exit ticket on a slip: *"Write the Sanskrit name and the principal aromatic compound of today's herb."*
- Restate the safety rule chorally.

- Preview Day 3: *"Tomorrow we meet a much less friendly tasting herb — neem."*

Student-facing content

Tulasī (*Ocimum tenuiflorum*) — known in English as holy basil, in Hindi as tulsi.

How to identify it: - Ovate (oval-pointed) leaf, about 2–5 cm long, with small jagged edges. - Square stem (a feature of the Lamiaceae family). - Strong, sharp, clove-like smell when you rub a leaf between your fingers.

What's inside: The main aromatic compound is **eugenol** — also the principal aromatic in actual clove. The plant produces it as a chemical defence.

Traditional uses: Kitchen herb (tulsi tea, chutney), household plant, ingredient in some Ayurvedic cold-and-cough preparations.

What we don't say: "Tulsi cures X." Folk use and preliminary evidence are not the same as clinical proof.

Homework

- Find a tulsi plant — your own pot, a neighbour's, a temple's, a school garden. Sketch its leaf in your workbook from life (not from a picture). Write one identifying feature you observed in person.

Differentiation

- **One tier up:** Sweet basil (*Ocimum basilicum*) has the same genus as tulsi but different uses. What chemical does sweet basil contain that tulsi doesn't dominate in (Hint: linalool)? Why does Italian cuisine use sweet basil but not tulsi?
- **One tier down:** Just remember the leaf shape and the word "eugenol." Smell-recognition will come with repetition.

Teacher notes

- If a student tells the class that tulsi is worshipped in their home, acknowledge respectfully: "Yes — many homes treat the tulsi plant as sacred. We're studying it here as a plant with specific chemistry; both framings can coexist."
- The clove-tulsi connection is the most memorable hook. Lean into it.
- Some students will react strongly to the smell of crushed tulsi (it can be quite sharp). Don't put it under their nose — let them choose their distance.

Citation(s) used in this lesson

- General phytochemistry: eugenol as principal volatile in *Ocimum tenuiflorum* — this is established botanical-chemistry textbook content, not a contested claim. No specific paper cited.

Day 3 — *Nimba* (Neem)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *nimba* (*Azadirachta indica*) by leaf shape, describe its dominant taste as *tikta* (bitter), name azadirachtin as its principal active compound, and explain why concentrated neem is **not** safe for ingestion at any age.

Materials

- A few dried neem leaves in a sealed container (smelling only — no chewing)
- A picture or — if you can — a neem twig with the characteristic compound leaf
- A small bottle of neem oil **kept by the teacher** (do not pass to students)
- A piece of neem-containing soap (commercial — most Indian general stores stock these) for the "what's it actually in" demonstration
- Whiteboard

Vocabulary introduced today

- *nimba* (IAST: nimba; lit. "the bitter one") — *Azadirachta indica*, the neem tree.
- *tikta* (IAST: tikta; lit. "bitter") — one of the six tastes in Ayurveda; neem is the canonical *tikta* plant.
- *azadirachtin* — a complex limonoid compound. The most-studied bioactive in neem; the molecule behind neem's use as an agricultural insect repellent.

Lesson flow

Warm-up (5 min)

Daily check-in: who has seen a neem tree near their home, school, or daily route? Hands.

"Anyone here ever been told 'eat a neem leaf, it's good for you'? Did you?" (Most who tried will say it tasted terrible.)

Bridge: today we explore why the body's *first response* to neem — "spit it out" — is actually telling us something useful.

Core (20 min)

1. **Identify it.** Neem is a tree, not a small herb:
    - **Compound leaf:** instead of a single blade, each "leaf" is made up of 9–15 small leaflets attached to a central stem. (Compare to tulsi yesterday — single simple leaves.)
    - **Leaflet shape:** narrow, curved (sickle-shaped), with serrate (toothed) edges.
    - **Smell when crushed:** strong, slightly garlic-like, mostly bitter.
    - **Where it grows:** native to South Asia; widely planted along streets, in temple compounds, and as a shade tree.
2. **The taste.** Pass the sealed container around for smell only.

   Today we do not chew the leaf.

Neem is the textbook example of *tikta* (bitter taste) in Ayurveda. The bitter taste comes from a class of compounds called **limonoids** — the most-studied of which is **azadirachtin**. Bitter taste is the body's warning signal — many plant toxins are bitter, and the bitter receptors evolved to make us *cautious* about new plants.

3. **What azadirachtin does.** This is real:

    - **Insecticidal/insect repellent:** azadirachtin disrupts insect moulting hormones. Hence neem oil is widely sold as an agricultural insect repellent — and that use is well-supported.
    - **Antimicrobial effects** at lab/petri-dish level.
    - **Traditional uses:** twigs as natural toothbrushes (*dātun*), neem-leaf paste on skin in some folk traditions, neem-based soaps and shampoos. These external/dilute uses are different from internal use.
4. **The safety warning — direct.** Write on the board:

    > ⚠️ **Neem is hepatotoxic at high doses.** Concentrated neem (especially neem oil ingested) has documented cases of acute liver injury, particularly in children. Neem oil is **not** for eating, drinking, or putting in anyone's mouth.

    This is a real, documented safety concern in case-report literature. Neem leaf in small amounts (a few leaves brewed as a bitter tea in some traditions) is generally tolerated by adults — but it is **not** what we are teaching here, and self-prescription is not appropriate.

5. **What's it in, that we DO use?**

    - Neem-based soap, shampoo, toothpaste — fine for external use as marketed.
    - Neem oil for plants — used as agricultural pest control. Read the label.
    - Neem twigs as toothbrushes — a long-standing dental tradition with modern dentistry's mixed view (some plaque-reducing benefit; abrasive enamel concern if used too often).

Activity (15 min) — Formative Quiz Part A + Compound-leaf sketch

First 5 min — Formative Quiz Part A (from `quizzes/formative.md`). Hand out the 5-question slip. Pencils only. Collect at minute 5.

Next 10 min — Compound-leaf sketch. - Each student draws a neem compound leaf from observation (or from the photo if no live sample). Label the rachis (central stem), the leaflets, the serrate edges. - Below the drawing, label: - **Sanskrit name:** *nimba* - **Botanical name:** *Azadirachta indica* - **Taste class:** *tikta* (bitter) - **Active compound:** azadirachtin - **Safety:** *not for ingestion in concentrated form*

Wrap-up (5 min)

- Restate safety rule + the neem-specific warning chorally.
- Hand-back of Day 2 exit tickets if marked.
- Preview Day 4: *"Tomorrow — the herb that turns everything gold."*

Student-facing content

Nimba (*Azadirachta indica*) — known in English as neem.

How to identify it: - A tree, not a small plant. - **Compound leaf** — what looks like one "leaf" is actually made of 9–15 small leaflets arranged along a central stem. - Leaflets are narrow, slightly curved, with jagged edges. - Smell when crushed: bitter, slightly garlic-like.

What's inside: Azadirachtin — a limonoid compound. The molecule behind neem's well-supported use as an agricultural insect repellent.

Taste class in Ayurveda: *tikta* — bitter. Bitter taste is the body's warning signal for many plant toxins.

Safety: - External uses (soap, oil for plants, twig as toothbrush) — fine when used as labelled. - **Concentrated internal use — especially neem oil — is unsafe.** Documented cases of liver toxicity in children. - We do not chew, drink, or eat neem in this class.

Homework

- Find a neem tree near your home or school. Sketch a leaf in your workbook from life. Note where the tree is and how many you saw on your walk.

Differentiation

- **One tier up:** Read up on the *Margosa oil poisoning* literature (margosa is another name for neem oil). Find one case-report finding. Be ready to share next class.
- **One tier down:** Focus on identifying the compound-leaf shape. The chemistry (azadirachtin) can come on Day 6 once you've seen more herbs.

Teacher notes

- The "don't chew the leaf" rule is critical. A few brave/curious students will try if leaves are accessible — keep the container sealed and store it where students can't reach during break.
- The hepatotoxicity warning needs to land as **specific knowledge**, not vague caution. Use the word "liver" explicitly. Some students have grandparents who give them neem leaves "for blood purification" — the lesson is to be respectful and curious about the traditional use *while* being clear about the dose concern.
- If a student asks "but if azadirachtin is dangerous, why have people used neem for centuries?" — the answer is: dilute external use and very small dietary use have a long history of being tolerated; *concentrated* use (oil, multi-leaf daily intake in adults, anything in children) is where the harm shows up.

Citation(s) used in this lesson

- *Azadirachta indica* and azadirachtin as principal limonoid: standard ethnobotanical and pharmacognosy textbook content.

- Neem oil hepatotoxicity: documented in published case reports in pediatric toxicology literature. We do not cite a specific paper here — teachers who want to deepen this should search “margosa oil hepatotoxicity case report” in PubMed.

Day 4 — *Haridrā* (Turmeric)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *haridrā* (*Curcuma longa*) as a rhizome (not a root), name **curcumin** as its principal pigment and active compound, distinguish whole-spice culinary use from concentrated supplement use, and describe one well-supported and one over-claimed use.

Materials

- Fresh turmeric rhizome (the orange-coloured fingerlike root-like piece) — ideally with a fresh piece cut to show the inside
- A small bowl of turmeric powder
- A piece of white paper for the staining demo
- A small piece of black pepper or a peppermill (for the curcumin-piperine pairing discussion)
- Optional, for the demo: warm milk, a small saucepan, turmeric powder, a pinch of pepper, jaggery — to make a single demonstration “golden milk” (subject to school food policy)
- Whiteboard

Vocabulary introduced today

- *haridrā* (IAST: haridrā; lit. “the yellow one”) — *Curcuma longa*, the turmeric plant.
- **Rhizome** — an underground horizontal stem (not a root). Distinguishes turmeric and ginger from carrot or beet.
- **Curcumin** — the polyphenol responsible for turmeric’s yellow colour and most of its studied biological activity.

Lesson flow

Warm-up (5 min)

Hold up the fresh turmeric rhizome. Slice a tiny piece — bright orange interior. Pass on a small plate so students can see.

“Hands up — who used turmeric in something you ate yesterday?” (Almost everyone.) *“Hands up — who knows that the bright yellow colour is one specific molecule?”* (Almost nobody.)

That’s today’s hook: a single molecule is responsible for almost everything turmeric is famous for.

Core (20 min)

1. **Identify it — rhizome, not root.** Important distinction:
    - A **root** absorbs water and nutrients (carrot, beetroot).
    - A **rhizome** is an underground stem that stores starch and produces new shoots (turmeric, ginger, ginger lily, bamboo).
    - Turmeric rhizomes are fingerlike, with skin that's pale brown to grey on the outside and bright orange on the inside.
    - The plant above ground has long, broad green leaves — similar to ginger and to canna lily.
2. **Powder vs whole.** Both forms are used:
    - **Fresh rhizome:** sliced into curries, pickled, sometimes juiced.
    - **Powdered (dried, ground):** the everyday Indian kitchen form — added to dal, sabzi, curries.
    - **Why the colour?** Curcumin and related curcuminoids — diarylheptanoid polyphenols. The molecule absorbs blue light and reflects yellow-orange.
3. **Staining demo (1 min).** Dab a small amount of fresh turmeric on white paper. Permanent stain. *"This is why turmeric on your shirt is a problem. Curcumin binds to fibres."*
4. **What modern chemistry knows.** Real, well-supported:
    - **Anti-inflammatory mechanism:** curcumin inhibits several inflammation-related signalling molecules in lab studies. Mechanism evidence is strong.
    - **Low bioavailability:** here's the catch — curcumin is **poorly absorbed** by the human gut on its own. Most of what you eat passes through unabsorbed.
    - **Piperine (black pepper) increases absorption** by roughly 20×, by inhibiting a liver enzyme that breaks down curcumin. This is why traditional preparations often combine turmeric with pepper. The tradition figured out the absorption problem long before the molecule was named.
5. **What's claimed vs what's solid.**
    - **Well-supported:** Turmeric is a safe culinary spice. It contributes to the colour and a mild flavour. Curcumin has anti-inflammatory activity in lab studies.
    - **Mixed evidence:** Curcumin supplements for arthritis pain — *some* clinical trials show modest benefit; others show no effect beyond placebo.
    - **Overclaimed:** "Turmeric cures cancer." This is a marketing claim, not a clinical finding. Curcumin is being studied for various cancer-related effects in the lab, but jumping from "lab activity" to "cure" is exactly the leap we are training students *not* to make.
6. **Traditional use in Ayurveda.** *Haridrā* is *kaṭu-tikta* (pungent-bitter) in taste, *uṣṇa* (heating) in *vīrya*. Used in formulations for skin conditions (external paste), wound healing (external paste), and digestive support (small culinary amounts). The bright pigment is also used at weddings (*haldi* ceremony) — non-medicinal cultural use.

Activity (15 min) — Golden-milk demonstration (or recipe-card if no cooking allowed)

If cooking is permitted: - Teacher prepares a single demo cup: 200 ml warm milk + ¼ tsp turmeric powder + tiny pinch black pepper + jaggery to taste. Warm gently (do **not** boil). - Don't drink in class. **The point is the recipe, not the consumption.** Or, prepare a cooled small tasting cup and offer to those without dairy/curcumin/dietary concerns.

If cooking is not permitted (or as a complement): - Each pair writes out a recipe card: ingredients, quantities, why each is there. - Specifically: *why pepper?* (bioavailability) *why warm not hot?* (palatable; also avoids breakdown) *why jaggery not white sugar?* (cultural preference; jaggery is unrefined cane sugar; no major chemistry difference for this purpose).

Either way: 5 minutes of write-up on a reflection slip — *"What did you learn today about claims around turmeric? Pick the most surprising thing."*

Wrap-up (5 min)

- Restate the safety rule. Add the turmeric-specific note: **turmeric supplements at multi-gram daily doses have caused liver issues in some individuals, particularly when combined with other supplements.** Culinary use in food is not the same as supplement use.
- Preview Day 5: *"Tomorrow, ginger — the kitchen rhizome that's also a key ingredient in many cold-and-flu home remedies."*

Student-facing content

> ***Haridrā*** (*Curcuma longa*) — known in English as turmeric, in Hindi as haldi.
>
> **What it is:** A **rhizome** — an underground stem, not a root. Bright orange inside; pale brown outside.
>
> **What's inside: Curcumin** — the polyphenol pigment. Same molecule that gives the yellow colour and most of the studied biological activity.
>
> **The bioavailability problem and the pepper fix:** - Curcumin is poorly absorbed by the gut on its own. - Black pepper (piperine) increases absorption by ~20×. - Traditional Indian cooking often pairs the two — the tradition got there empirically.
>
> **Honest evidence summary:** - Safe and useful as a culinary spice. - Strong lab evidence for anti-inflammatory mechanism. - **Mixed** clinical evidence for specific health claims; **no** evidence supports "cancer cure" claims.
>
> **Traditional Ayurvedic classification:** *kaṭu-tikta* (pungent-bitter) in taste, *uṣṇa* (heating) in potency.

Homework

- Find turmeric in your home (powder, fresh, capsules, golden milk — anything). Note: where is it? What form? Has anyone in your family ever said "turmeric helps with X"? If yes, write the claim down — we'll look at it on Day 8.

Differentiation

- **One tier up:** Curcumin is fluorescent under UV light. Look up *one* other dietary pigment that fluoresces. Why might fluorescence and biological activity be related (hint: think about how light energy interacts with electronic structure)?
- **One tier down:** Focus on identifying the rhizome shape and remembering the word "curcumin." Stain demo and bioavailability concept come back on Day 10's revision.

Teacher notes

- **Stain risk in the classroom is real.** Cover desks. Have warm water + soap nearby. A fresh curcumin stain on cloth washes out before it sets; once it sets and gets sun, it sticks.
- The bioavailability + black pepper story is the **single biggest "aha"** of this module for many students. Their families may have known to use the two together without knowing why. Let students go home and ask.
- Don't let the "cancer cure" myth go unaddressed. It's everywhere on social media. Be specific: "lab activity ≠ clinical effect ≠ cure."

Citation(s) used in this lesson

- Curcumin chemistry and bioavailability with piperine: well-established in nutrition and ethnopharmacology textbooks. Modern reviews exist in journals such as *Foods*, *Nutrients*, and the *Journal of Ethnopharmacology* — we do not cite a specific paper because RAG verification wasn't available; teachers can search "curcumin piperine bioavailability review" in any indexing database.
- Turmeric as *haridrā* in Ayurveda — standard reference, *Caraka Saṁhitā Sūtrasthāna* (chapter level only).

Day 5 — *Adraka* (Ginger)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *adraka* (*Zingiber officinale*) as a rhizome, distinguish *adraka* (fresh) from *sūnṭhī* (dried) and explain that the active compound shifts from **gingerol** to **shogaol** on drying, and describe ginger's traditional classification as *uṣṇa* (heating) with one well-supported modern use.

Materials

- Fresh ginger rhizome
- Small jar of dried ground ginger powder (*sūnṭhī*)
- Optional: a thumb-size lump of jaggery for the heating-demo
- Optional, for demo: a small kettle, ginger slices, water, lemon, jaggery — ginger tea
- Whiteboard

Vocabulary introduced today

- *adraka* (IAST: adraka; lit. “ginger”) — fresh ginger rhizome.
- *sūṇṭhī* (IAST: sūṇṭhī; lit. “dried ginger”) — same plant, dried rhizome.
- **Gingerol** — the principal pungent compound in fresh ginger ([6]-gingerol).
- **Shogaol** — formed when gingerol loses a water molecule during drying or heating. More pungent. The dominant pungent compound in dried ginger.

Lesson flow

Warm-up (5 min)

Hold up the fresh ginger and the dried ginger powder.

“Same plant. Same rhizome. One fresh, one dried. Which one will be more pungent?” Take a vote.

Pass tiny crumb of dried ginger and a tiny shaving of fresh ginger to each student (separate plates). Smell only. Discuss.

Core (20 min)

1. **Identify it.** Like turmeric, ginger is a **rhizome** — underground stem. The two plants are related (same family: Zingiberaceae). Differences:
    - **Ginger:** pale tan skin; pale yellow-white interior; spicy-pungent smell.
    - **Turmeric:** pale brown skin; bright orange interior; earthy-pepper smell.
    - **Leaves above ground:** very similar — long, blade-like, alternating up a stem.
2. **The fresh-vs-dried switch.** This is the chemistry of the day.
    - **Fresh ginger** is dominated by **[6]-gingerol** — a phenolic compound with a long carbon side-chain ending in a hydroxyl (–OH).
    - When ginger is **dried** (loses water content) or heated, gingerol undergoes a dehydration reaction. The hydroxyl group is lost as water, leaving **[6]-shogaol**.
    - Shogaol is more pungent than gingerol — it’s why dried ginger powder is sharper on the tongue than fresh ginger.
    - In Ayurveda, this same transformation is noticed: *adraka* (fresh) is described as somewhat less heating than *sūṇṭhī* (dried). The classical text and the modern molecule agree.

    Sketch on board:

    ```
    gingerol  →  (drying / heating)  →  shogaol  +  H₂O
    (fresh)                            (dried, more pungent)
    ```
3. **What ginger does — well-supported uses.**
    - **Anti-nausea** — one of the few traditional herbs with **multiple positive clinical trials**, especially for:

        - Pregnancy-related nausea (under doctor supervision — pregnancy dosing is not a school question).
        - Motion sickness — small ginger doses (1–2 g, e.g. a chewed slice) have outperformed placebo in some trials.
        - Post-surgical nausea — used in some hospital protocols.
    - **Warming feeling** — ginger does cause a measurable rise in skin and body temperature (“thermogenic” effect). Activates TRPV1 receptors — the same receptor activated by capsaicin in chillies.
    - **Digestive support** — culinary use after meals (a slice in tea, or ginger pickle) has a long tradition; mechanism includes stimulation of saliva and gastric secretions.
4. **Traditional Ayurvedic frame.** *Adraka* is *kaṭu* (pungent) in taste, *uṣṇa* (heating) in *vīrya*. Used in:
    - Tea / decoction for cold, congestion, light fever (small culinary amounts).
    - Combined with jaggery for a sore throat (folk remedy; some mechanistic justification through the soothing/warming pairing).
5. **Cautions:**
    - Large doses can cause heartburn or stomach upset.
    - Pregnancy — doses higher than culinary need supervision.
    - Some blood-thinning medications (warfarin) interact with high ginger intake. Not a school-dose concern but worth knowing.

Activity (15 min) — Ginger tea demo + Formative Quiz Part B

First 7 min — Tea demo: - Teacher slices a 1 cm piece of fresh ginger, drops into 200 ml hot water, lets steep 3–4 minutes. - Optional: add a tiny pinch of jaggery and a drop of lemon. - Pour into a clear cup. Pass around so students can see the colour (very pale yellow) and smell the aroma (sharp, pungent, citrus-warm). - Discussion: which compound dominates here — gingerol or shogaol? (Since the ginger is fresh and only briefly heated: mostly gingerol.) If the ginger had been dried first and then steeped: more shogaol.

Next 8 min — Formative Quiz Part B (from `quizzes/formative.md`). Hand out the 10-question slip. Pencils only. Collect at end of class.

Wrap-up (5 min)

- Restate safety rule + the anti-nausea note (“ginger has more clinical evidence than most herbs, but for *specific* uses”).
- Preview Day 6: *“Tomorrow we meet a seed that takes the digestive role even further — ajwain.”*

Student-facing content

> ***Adraka*** (*Zingiber officinale*) — known in English as ginger.
>
> **What it is:** A **rhizome** (underground stem), related to turmeric. Pale tan outside, pale yellow inside.

Fresh vs dried — same plant, different chemistry:

Form	Sanskrit name	Dominant compound
Fresh rhizome	*adraka*	**gingerol**
Dried rhizome	*sūnṭhī*	**shogaol** (more pungent)

When ginger dries or heats, gingerol loses a water molecule and becomes shogaol.

Well-supported modern uses: - Anti-nausea, especially motion sickness and pregnancy-related (under medical supervision). - Mild warming effect (activates TRPV1 receptors — same as chilli's capsaicin). - Digestive support in culinary amounts.

Ayurvedic classification: *kaṭu* (pungent) in taste, *uṣṇa* (heating) in *vīrya*.

Cautions: Large doses → heartburn or stomach upset; pregnancy doses need a doctor; interactions with blood-thinning medications.

Homework

- Watch what ginger is used in your home over the next two days. List 3 dishes / drinks that contain it. For each, guess: fresh or dried? Heating or just flavour?

Differentiation

- **One tier up:** Look up the chemical structure of gingerol and shogaol. Identify the difference (one OH group + one double bond gain). Sketch both in your notebook.
- **One tier down:** Focus on identifying ginger by sight and smell, and remembering "fresh = gingerol, dried = shogaol."

Teacher notes

- The chemistry sketch (gingerol → shogaol) is the headline. If a student remembers nothing else from today, they should know that drying ginger changes its chemistry, and the change is well-defined.
- The "ginger has real clinical evidence for nausea" point is important — students often don't realise that for *some specific herbal uses*, the clinical evidence is genuinely strong. It builds the right kind of nuance: not blanket scepticism, not blanket endorsement, but herb-by-herb, use-by-use evaluation.
- If a student in your class is pregnant (older Middle band, occasionally) or has a pregnant family member raising questions, redirect gently: clinical dosing is for clinicians.

Citation(s) used in this lesson

- Gingerol → shogaol dehydration on drying: standard pharmacognosy textbook content.
- Ginger as *adraka* (fresh) and *sūnṭhī* (dried) — distinguished in *Caraka Saṁhitā* and standard Ayurvedic nighantus (lexicons). Chapter-level reference only.
- Anti-nausea clinical evidence: multiple systematic reviews exist; teachers can search "ginger nausea systematic review" in PubMed.

Day 6 — *Yavānī* (Ajwain / Carom)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *yavānī* (*Trachyspermum ammi*) by seed appearance and crushed-seed smell, name **thymol** as the principal volatile, and describe one common culinary use linked to traditional digestive support.

Materials

- Small jar of whole ajwain seeds (one per pair, plus one display jar)
- Pestle and mortar (if available)
- A few sprigs of fresh ajwain leaf (*ajwain ka patta* — sometimes called Indian borage / Cuban oregano — different plant but commonly confused; clarify if you bring it)
- Optional: dried thyme (the European cousin chemistry-wise) for the smell-bridge
- Small bowl of warm water + a teaspoon of ajwain for the demo
- Whiteboard

Vocabulary introduced today

- *yavānī* (IAST: yavānī; lit. "barley-like") — the Ayurvedic name for ajwain. Also called *ajavāyana* in some texts.
- **Thymol** — the principal volatile aromatic in ajwain seed. Same compound dominates thyme (*Thymus vulgaris*).

Lesson flow

Warm-up (5 min)

Hold up the jar of ajwain seeds. Pour a small amount into a tray.

"What do these look like?" (Tiny, oval, brown-grey, ribbed.) *"What do you smell?"* (Sharp, warm, oregano-like / thyme-like.)

Pass a small pile to each pair. They crush a few seeds with the back of a spoon (no chewing yet) and sniff. The smell intensifies dramatically when crushed — release of volatile thymol.

Core (20 min)

1. **Identify it.**

    - *Yavānī* is a small annual herb in the Apiaceae family (same family as coriander, cumin, fennel, dill).
    - The "seeds" sold in shops are technically **fruits** (small schizocarps), about 2 mm long, oval, ridged, greyish-brown.
    - Crushed, they release a strong thyme-like aroma. The smell is the chemistry — thymol is volatile, so crushing exposes more of it.

2. **The ajwain-leaf confusion (1 min).** What's sold in Indian markets as "ajwain leaf" or "ajwain ka patta" is often *Plectranthus amboinicus* — Cuban oregano. It's a different plant, but it smells similar (both contain thymol-class compounds). Modern botany distinguishes them; in many kitchens they get used somewhat interchangeably. Note for clarity but don't dwell.

3. **What's inside.** Thymol is a **phenolic monoterpenoid**, $C_{10}H_{14}O$ — closely related to carvacrol. Properties:

   - **Antimicrobial** in vitro — used historically in mouthwashes and antiseptic preparations (Listerine famously included thymol).
   - **Carminative** — helps reduce intestinal gas. Mechanism likely involves smooth-muscle relaxation in the gut.
   - **Strong volatile** — that's why a tightly closed spice jar holds the smell, and an open one fades in a few weeks.
4. **Traditional use — culinary, not medicinal.**

   - Tempered into hot oil at the start of cooking parathas, kachoris, savoury snacks.
   - **Ajwain water** — a common Indian household remedy: 1 tsp ajwain steeped in 200 ml hot water for 5–10 minutes, strained, drunk warm. Used for indigestion, bloating, mild stomach discomfort.
   - Often added to flour-based foods that are heavy on starch (parathas, samosas) — the carminative effect helps with the digestive load.
5. **Honest evidence summary.**

   - **Mechanism evidence:** thymol relaxes gut smooth muscle and has antimicrobial action. Well established at the lab level.
   - **Clinical evidence:** for *general indigestion / dyspepsia*, the clinical trial picture is small but generally positive for thymol-containing preparations.
   - **Cautions:** essential oils made from ajwain are extremely concentrated — multi-fold more thymol per drop than the whole spice. **Not for ingestion at school age.** The whole seed in food is a different scale entirely.
6. **The doṣa frame.** *Yavānī* is *kaṭu* (pungent) and *tikta* (bitter), *uṣṇa* (heating). Considered to balance *vāta* and *kapha* (mainly by stimulating digestion).

Activity (15 min) — Ajwain-water demo + Compare to thyme

Demo (5 min): - Teacher places 1 tsp ajwain in a clear glass. Pours 200 ml hot (not boiling) water over it. Covers with a saucer. - Pass around the glass — observe the seeds float, the water gain colour, the smell rise. - 5 minutes later, uncover. The aroma is now intense. - This is the home remedy: strain and drink warm (for the demo, don't drink — just observe).

Compare to thyme (5 min): - Pass around dried thyme. Crush a small amount, smell. - The two herbs are unrelated botanically (different families) but share the same dominant volatile — **thymol**. This is a beautiful example of *convergent chemistry* — different plants evolving the same compound because it serves a similar purpose for the plant (defence against microbes and herbivores).

Exit ticket (5 min): - On a slip: *"Name today's herb in Sanskrit, its main compound, and one common Indian dish you've eaten it in."*

Wrap-up (5 min)

- Restate safety rule + the essential-oil warning ("the whole seed in food is fine; the essential oil is concentrated and not for school-age ingestion").
- Preview Day 7: *"Tomorrow — mint. The cooling herb."*

Student-facing content

> ***Yavānī*** (*Trachyspermum ammi*) — known in English as ajwain or carom seed.
>
> **What it is:** A small annual herb in the same family as coriander, cumin, and fennel (Apiaceae). The "seeds" you buy are technically tiny fruits.
>
> **How to identify:** Oval, ridged, greyish-brown, about 2 mm long. Strong thyme-like smell — much stronger when crushed.
>
> **What's inside: Thymol** — a phenolic monoterpenoid. Same dominant compound as European thyme — *convergent chemistry* between unrelated plants.
>
> **Traditional use:** Tempered into hot oil for parathas, savoury snacks. Ajwain water (steeped seeds in warm water) is a common Indian household remedy for indigestion.
>
> **Honest evidence:** Mechanism for carminative (anti-gas) action is well established. Clinical evidence for general indigestion is positive but small. **Essential oils are concentrated and not safe for ingestion in school-age use.**
>
> **Ayurvedic frame:** *kaṭu-tikta* (pungent-bitter), *uṣṇa* (heating). Balances *vāta* and *kapha* via digestive stimulation.

Homework

- Find ajwain in your kitchen. Smell it. Then ask one adult: *"What do you put ajwain in, and why?"* Write down their answer.

Differentiation

- **One tier up:** *Convergent chemistry* — find one other example of two unrelated plant families producing the same compound (e.g. cinnamaldehyde in cinnamon and in some flowers; geraniol in geranium, lemongrass, rose). Sketch the molecular structure of thymol if you can.
- **One tier down:** Focus on identifying ajwain seeds by smell and naming the compound "thymol."

Teacher notes

- The **crushed-seed smell intensification** is the most memorable hands-on of the day. Make sure every student gets to crush their own seeds. The "oh wow" reaction is the lesson.
- The Indian-borage / ajwain-leaf confusion is real and you'll get questions. Acknowledge — "the leaf you find in shops is from a different plant that smells similar" — but don't make the module about taxonomy.

- The ajwain-water demo is one of the safer cooking-adjacent demos. Even so: don't have students drink from a shared glass. Demonstrate; don't distribute.

Citation(s) used in this lesson

- Thymol as principal volatile in *Trachyspermum ammi*: standard ethnopharmacology / phytochemistry reference.
- Carminative mechanism of thymol: well-established at lab level; clinical evidence reviewed in standard ethnopharmacology journals. No specific paper cited.
- *Yavānī* in Ayurveda — standard nighantu references; chapter-level only.

Day 7 — *Pudīnā* (Mint)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *pudīnā* (*Mentha* species) by leaf shape and the characteristic cooling smell, name **menthol** as the principal volatile, distinguish two common varieties (spearmint vs peppermint), and explain that mint is classified as *śīta* (cooling) in Ayurveda — a classification that aligns with its measurable physiological effect.

Materials

- Two sprigs of mint, ideally from two different varieties (spearmint and peppermint are easiest to source)
- A small jar of dried mint leaves
- A few peppermint candies or a small tube of menthol-containing balm — for the smell bridge (do not let students apply balm to their skin in class)
- Ice cube + a small bowl (for the demo on cooling sensation)
- Whiteboard

Vocabulary introduced today

- *pudīnā* (IAST: pudīnā) — mint. Commonly refers to *Mentha spicata* (spearmint) in Indian kitchens.
- **Menthol** — the principal volatile cooling compound. Most concentrated in peppermint (*Mentha* × *piperita*); also present in smaller amounts in spearmint.
- *śīta* (IAST: śīta; lit. "cool, cooling") — the cooling potency (*vīrya*) in Ayurveda. Contrasts with *uṣṇa* (heating).

Lesson flow

Warm-up (5 min)

Pass the mint sprigs along the row. Each student takes a leaf, rubs gently, sniffs.

"Three words for the smell." (Likely: "fresh," "cool," "toothpaste-like.")

Hold an ice cube briefly — note how cold feels on the skin. Now ask: *"When you smelled the mint just now, did it feel like the ice or different?"* (Different — but related. We'll explain.)

Core (20 min)

1. **Identify it.**
    - **Mint family (Lamiaceae)** — same family as tulsi (square stems!). Make students check: press the mint stem between thumb and finger. Four flat sides.
    - **Leaf shape:** oval-pointed, serrate (jagged) edges. Larger than tulsi leaves. Bright green.
    - **Smell:** the unmistakable cooling-fresh aroma. Different varieties smell slightly different.
    - **Common varieties in India:**
        - *Mentha spicata* — spearmint. The everyday "pudina" in chutney, raita.
        - *Mentha* × *piperita* — peppermint. More concentrated menthol; sharper, "tooth-paste" smell.
        - *Mentha arvensis* — corn mint / Japanese mint — grown commercially in India for menthol extraction.
2. **The cooling trick — explained.** This is the headline.
    - Menthol activates a receptor in your skin called **TRPM8** — the same receptor that activates when your skin actually gets cold.
    - The receptor doesn't know whether the signal is from real cold or from a menthol molecule. It sends "cold!" to your brain either way.
    - The result: **you feel cool without your temperature actually dropping**. Your skin temperature is unchanged. Your perception of coolness is real.
    - This is the same trick that capsaicin (chilli) plays in reverse — capsaicin activates TRPV1, the "heat" receptor, and your brain feels heat even though your skin isn't hot.

    This is one of the cleanest connections between **traditional classification** (*śīta* — cooling) and **modern molecular pharmacology** (TRPM8 activation). The tradition got the *effect* right; modern science explained the *mechanism*.

3. **What mint does — well-supported.**
    - **Cooling sensation** (TRPM8) — definitively established.
    - **Digestive support** — peppermint oil capsules have **good clinical evidence** for relieving symptoms of Irritable Bowel Syndrome (IBS) when prescribed by a doctor. Mechanism: relaxation of gut smooth muscle.
    - **Breath freshness** — antibacterial activity in the mouth.
    - **Headache relief** when applied to the temples (the cooling sensation distracts from pain). Topical use of dilute menthol; not the same as ingesting essential oil.
4. **Cautions.**

    - **Babies and small children** — undiluted menthol-containing products on the chest or face can cause breathing reflexes. Do not give menthol balms to infants.
    - **Peppermint oil capsules** — these are *medications*, not snacks. The IBS evidence is for *prescribed* dosing.
    - **Acid reflux** — peppermint can relax the lower oesophageal sphincter, making reflux worse for some people.
5. **Traditional Ayurvedic frame.** *Pudīnā* is *kaṭu* (pungent) in taste but *śīta* (cooling) in *vīrya*. Used in summer drinks (mint nimbu pani), in chutneys (with coriander and lime), and in some preparations for cooling *pitta* imbalances.

Activity (15 min) — Variety taste-id challenge

(Skip if cooking/tasting is not permitted at your school. Replace with a sketch-by-variety exercise.)

- Set up two cups: spearmint chutney (from prepared chutney, very small portion per student) vs a peppermint candy crushed in water (tiny amount).
- Each pair tries to identify which is which by taste alone.
- Record what made them distinguishable — sharpness? Sweetness? Cooling intensity?

Final 5 min — group discussion: - *"Why does mint feel cool when it isn't actually cold?"* (TRPM8 activation — students should be able to explain this back.) - *"What's the connection between the Ayurvedic word śīta and this molecular trick?"* (The tradition was tracking the effect; the science explains the cause.)

Wrap-up (5 min)

- Restate the safety rule.
- Preview Day 8: *"Tomorrow we put all six herbs together — and look at them through the doṣa lens. Heating, cooling, what goes with what."*

Student-facing content

> ***Pudīnā*** (*Mentha* species) — mint.
>
> **What it is:** A Lamiaceae herb (square stems!). Common varieties in India: *Mentha spicata* (spearmint), *Mentha × piperita* (peppermint), *Mentha arvensis* (corn mint).
>
> **How to identify:** Oval-pointed serrate leaves, bright green; the unmistakable cooling-fresh smell when crushed.
>
> **What's inside: Menthol** — the principal volatile.
>
> **Why does mint feel cool?** - Menthol activates **TRPM8** — the same receptor in your skin that activates when you actually feel cold. - Your brain receives the "cold!" signal even though your skin's temperature hasn't changed. - This is the **mechanism** behind the Ayurvedic classification of mint as *śīta* (cooling) — the tradition observed the effect; modern pharmacology explained the cause.

Well-supported modern uses: - Cooling sensation (mechanism: TRPM8 activation). - Peppermint oil for IBS symptoms (prescribed, not self-administered). - Antibacterial action in the mouth (breath freshness).

Cautions: Not for infants on the chest/face. Concentrated essential oil and capsules are medications, not snacks.

Homework

- Find pudina (or any mint) in your home or near it. Sketch the leaf. Identify which variety if you can. Note: where do you encounter mint most often in your week?

Differentiation

- **One tier up:** TRPM8 was identified as the cold receptor in 2002. Look up the David Julius lab work on TRP channels (2021 Nobel Prize in Physiology or Medicine). Be ready to share one finding next class.
- **One tier down:** Focus on identifying mint by smell and remembering “menthol = cool feeling.”

Teacher notes

- The **TRPM8 mechanism** is the single most rewarding teaching moment of this module for a science-leaning student. The bridge from “cooling herb” (tradition) to “TRPM8 activation” (Nobel-winning science) is satisfying and intellectually honest.
- If anyone in your class connects “menthol cooling = chilli burning = same kind of receptor trick,” that’s the inferential leap to highlight publicly. Praise it.
- A few students may bring up vapour rubs / chest balms used in childhood. Acknowledge the use; note the infant caution; don’t sermonise.

Citation(s) used in this lesson

- Menthol activation of TRPM8: established in molecular pharmacology. David Julius and colleagues identified TRPM8 as the cold/menthol receptor (Julius lab, UCSF; 2021 Nobel Prize work). Reference is at the lab level only; specific paper not cited.
- Peppermint oil for IBS: multiple systematic reviews exist. Teachers can search “peppermint oil IBS systematic review” in PubMed.
- *Pudīnā* as *śīta* — standard Ayurvedic nighantu reference; chapter-level only.

Day 8 — The Doṣa Lens: Heating and Cooling

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can sort all six module herbs into heating (*uṣṇa*) and cooling (*śīta*) categories, explain the Ayurvedic concept of *vīrya* in one sentence, link this classification to specific *doṣa* imbalances from Module 4, and identify one example where the traditional classification matches a measurable modern pharmacological effect.

Materials

- A printed 6-card set with each herb (one per pair); these can be drafts of the project deck
- Whiteboard with two columns: **uṣṇa** (heating) | **śīta** (cooling)
- Sticky notes
- Optional: Day 4–7 worksheets students completed, so they have their own data

Vocabulary introduced today

- *vīrya* (IAST: vīrya; lit. "potency, energy") — in Ayurveda, the heating or cooling *potency* of a substance, distinct from its taste (*rasa*) and its post-digestive effect (*vipāka*). At Middle band we use only the heating/cooling axis.
- *uṣṇa* (IAST: uṣṇa; lit. "hot, heating") — the heating *vīrya*.
- *śīta* (IAST: śīta; lit. "cold, cooling") — the cooling *vīrya*.

Lesson flow

Warm-up (5 min)

Daily prompt: *"Which herb did you smell or eat at home since our last class?"* Two students share for 30 sec each.

Now the day's hook: *"From what you remember about each herb — which feels 'warming' when you eat it? Which feels 'cooling'?"* Take a few intuitions, write them on the board.

Core (20 min)

1. **What is *vīrya*?** Write on board:

   > *vīrya* — the **heating** or **cooling** potency of a substance, measured by its effect on the body.

   This is one of four properties Ayurveda uses to classify any food or herb:

   - *rasa* — taste (sweet, sour, salty, pungent, bitter, astringent)
   - *vīrya* — heating or cooling potency
   - *vipāka* — post-digestive effect
   - *prabhāva* — special, idiosyncratic action

   At Middle band, we focus on *vīrya*. The full fourfold framework comes back at Senior band.

2. **Sort the six herbs.** Build the two-column table on the board, asking students to predict before you confirm:

uṣṇa (heating)	*śīta* (cooling)
tulasī — eugenol, warming aromatic	*pudīnā* — menthol, TRPM8 cooling
haridrā — turmeric, mild warming	
adraka — ginger, strongly warming (TRPV1)	

uṣṇa (heating)	*śīta* (cooling)
yavānī — ajwain, warming digestive	
nimba — neem (slightly contested — see below)	

The contested case: *nimba* (neem) is classified as bitter (*tikta*) and *śīta* (cooling) in classical Ayurveda — even though it's a bitter herb without an obviously cooling sensation. Why? Because in the *doṣa* model, bitterness reduces *pitta* (the "fire" doṣa) — and a substance that reduces *pitta* is by definition cooling in the system. Bring this up explicitly to show students that the Ayurvedic categories are **functional**, not always sensory.

3. **Why does heating/cooling matter?** Link back to *doṣa*s (refresh briefly if Module 4 hasn't been taught):

    - **Vāta** (motion, air-and-space) — generally treated with warming, moistening, grounding foods/herbs. Cold and dry aggravate vāta.
    - **Pitta** (transformation, fire-and-water) — treated with cooling, mild foods. Heat and pungency aggravate pitta.
    - **Kapha** (structure, water-and-earth) — treated with warming, light, drying foods. Cold and heavy aggravate kapha.

    So:

    - **Ginger + ajwain** → useful for *kapha* and *vāta* imbalances (warming, stimulating).
    - **Mint** → useful for *pitta* imbalances (cooling).
    - **Tulsi** → balanced for vāta/kapha (warming, but not as strongly as ginger).
    - **Turmeric** → mildly heating but broadly used because of its antimicrobial profile.
    - **Neem** → useful for *pitta* and *kapha* imbalances (its bitterness reduces both, in the framework's logic).

4. **The match between tradition and modern pharmacology.** This is the most important slide of the module:

Herb	Tradition says	Modern says
pudīnā (mint)	*śīta* — cooling	Menthol activates TRPM8 → "cool" signal to brain
adraka (ginger)	*uṣṇa* — heating	Gingerol activates TRPV1 → "warm" signal to brain

Both classifications were made by careful observers — millennia apart, using completely different methods. They agree because they were both tracking a real biological effect.

5. **Source — paraphrase of the Auṣadhi-sūkta** (Ṛgveda 10.97). Read aloud once:

    "O herbs, born before the gods, brown and bright, born of past ages — may you who carry many forms protect the one who calls on you."

    — paraphrase of the opening of Ṛgveda 10.97, the *Auṣadhi-sūkta*

> Comment: the hymn is **3000+ years old**. The plants are addressed as agents — like persons. Whether or not you take that literally, notice what it does: it treats the plant as something to **pay attention to**. The *doṣa* framework is built on millennia of paying that kind of attention.

Activity (15 min) — Sort by vīrya (see `activities/activity-02-sort-by-virya.md`)

Pairs receive a set of 12 cards: the 6 module herbs + 6 other common kitchen items (rice, ghee, lemon, ice cream, chilli, coriander seed). They sort into *uṣṇa* / *śīta* / *not sure* and defend their reasoning.

Last 3 min: each pair shares their **most uncertain** card and why.

Wrap-up (5 min)

- Restate safety rule.
- Hand out the project deck instructions for Day 9. Show the rubric on the board. Walk through one card's expected content.
- Preview Day 9: *"Tomorrow you build your six-card herb deck. Bring your sketches, your notes, your colour pencils."*

Student-facing content

> **Vīrya** is the heating or cooling *potency* of a substance in Ayurveda.
>
> **Heating (*uṣṇa*) — tulsi, turmeric, ginger, ajwain. Cooling (*śīta*) — mint. Neem too — though in a functional, not sensory, way.**
>
> **Why does this matter?** - Heating herbs balance *vāta* and *kapha* imbalances (cold-related, sluggish conditions). - Cooling herbs balance *pitta* imbalances (heat-related, inflammatory conditions).
>
> **Two beautiful matches between tradition and modern science:** - *Pudīnā* is *śīta* — menthol activates TRPM8, the same skin receptor that fires when you actually feel cold. - *Adraka* is *uṣṇa* — gingerol activates TRPV1, the same skin receptor that fires when you actually feel heat.
>
> Both classifications were made by careful observers, millennia apart, using completely different methods. They agree because they were tracking a real biological effect.
>
> "O herbs, born before the gods, brown and bright, born of past ages — may you who carry many forms protect the one who calls on you." — paraphrase of the opening of the *Auṣadhi-sūkta* (Ṛgveda 10.97)

Homework

- Finalise your project pair. Decide which 2 herbs you'll each draft cards for first (you'll do 3 each). Bring everything tomorrow.
- One sentence answer for the next class: *"What is the most surprising thing you've learned about herbs in the past week?"*

Differentiation

- **One tier up:** The full Ayurvedic fourfold framework is *rasa-vīrya-vipāka-prabhāva*. Look up what *vipāka* means and find one example where a herb's *vipāka* differs from its *rasa*.
- **One tier down:** Focus on the heating/cooling sort. The *doṣa* connection comes back in summative review.

Teacher notes

- The TRPM8/TRPV1 ↔ *śīta*/*uṣṇa* convergence is the cleanest "tradition and modern science describe the same biological reality" moment in the module. Lean into it. Don't oversell ("Ayurveda discovered TRP channels first" — no, they didn't, that's a different question) — but don't undersell either.
- The neem-is-cooling-because-bitter point will surprise students. Use it to discuss what "potency" means in a functional framework vs a sensory one.
- The Auṣadhi-sūkta paraphrase is *not* a translation. Don't pass it off as a quote. Be explicit: this is a paraphrase of the opening of Ṛgveda 10.97. Teachers who want a published translation should consult Griffith (1896) or Jamison & Brereton (2014).

Citation(s) used in this lesson

- *Auṣadhi-sūkta* — Ṛgveda 10.97 — referenced and paraphrased. Public-domain Vedic source.
- *Caraka Saṁhitā* — *vīrya* as one of four properties of substances. Chapter-level reference to *Sūtrasthāna*.
- TRPM8 / TRPV1 work — established neuroscience. 2021 Nobel Prize in Physiology or Medicine to David Julius and Ardem Patapoutian.

Day 9 — Project Session: Build the Herb Deck

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, each pair has produced **all six cards** in draft form, each containing the eight required elements (see below). The cards are ready for finalising at home and presenting on Day 10.

Materials

- Pre-cut A6 (or quarter-A4) cards — 6 per pair, plus 2 spares
- Colour pencils, fine-tip markers, scale rulers
- Reference: students' Day 2–8 worksheets (instruct them to bring these)
- Project brief (`assessments/project-brief.md`) printed and visible
- Rubric (`assessments/rubric.md`) printed and visible

Lesson flow

Warm-up (5 min)

Read the project brief aloud (one student volunteers).

Restate the **eight required elements** on each card: 1. Sketch of the herb (leaf, root, or seed — whichever is most identifying) 2. **Sanskrit name** in IAST 3. **English / common name** 4. **Botanical name** (italicised) 5. **Family** (e.g. Lamiaceae, Apiaceae, Zingiberaceae) 6. **Three identifying features** (smell, leaf shape, colour, etc.) 7. **Active compound** (curcumin, eugenol, etc.) 8. **One traditional use + one safety note**

Show one finished card example (you can use yesterday’s *pudīnā* card-template).

Core / activity (35 min)

Pairs work continuously. The teacher rotates and intervenes.

Pacing checkpoints — write on the board:

- **By minute 12 (out of 35):** All six cards should have the sketch + Sanskrit + English name. (The fastest part.)
- **By minute 25:** All six cards should have the active compound and one traditional use.
- **By minute 35:** Each card should have the safety note + three identifying features.

If a pair lags at any checkpoint, intervene specifically — don’t let them spend 15 minutes perfecting one card.

Quality interventions to watch for: - Cards copying directly from a worksheet without engagement → ask “what’s the most interesting thing about this herb? Add that to the card.” - Cards with no safety note → “what’s the dose question for this herb?” Write the answer. - Cards with grand health claims (“turmeric cures arthritis”) → push back: “what does the evidence actually say?” - Cards missing the family → look it up together on the board (Lamiaceae for tulsi+mint, Apiaceae for ajwain, Zingiberaceae for ginger+turmeric, Meliaceae for neem).

Wrap-up (5 min)

- Each pair holds up one card they’re proudest of. 10-second show + tell.
- Collect cards in pair-folders for tomorrow. (Or have pairs take them home if they prefer to finalise overnight.)
- Hand out the reflection slip — to complete tonight as homework: > *“What I set out to learn, what I actually learned, and one question I still have.”*

Student-facing content

> **Today: build your 6-card herb deck.**
>
> Each card has eight things — sketch, Sanskrit name, English name, botanical name, family, three identifying features, active compound, and one traditional use + one safety note.

Quality over completeness — but all six cards must be drafted by end of class.

Use what you've already drawn and noted across Days 2–7. Your worksheets are your raw material; the cards are the polished version.

Homework

- Finalise any card you didn't complete in class.
- Complete the reflection slip.
- Bring all six cards + reflection slip + pencils to Day 10.

Differentiation

- **One tier up:** Make a **7th card** for a herb of your choice — *aśvagandhā*, *brāhmī*, *triphalā*, *kariveppilai* (curry leaf), or *dhānyaka* (coriander). Apply the same eight-element template.
- **One tier down:** Make 4 cards (drop turmeric and ajwain if compressed) — but each must have all 8 elements. Quality over quantity.

Teacher notes

- This is a build session, not a teaching session. Resist the urge to lecture. Walk around. Ask questions. Point at gaps.
- Common failure mode: pairs spend the first 15 min discussing what to draw and end up with one half-card. Set the timer visible. Push pace.
- Common failure mode: one partner draws while the other watches. Reassign — partner A does 3 cards in pen, partner B does 3 cards in pen, then they swap and refine each other's.

Citation(s) used in this lesson

- Project brief — `assessments/project-brief.md`
- Rubric — `assessments/rubric.md`

Day 10 — Showcase + Summative Quiz

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

Students present their herb deck to the class (2 minutes per pair), complete the summative quiz (20 min, 25 marks), and consolidate the module's safety message and tradition-meets-science framing.

Materials

- All pairs' completed decks
- Printed copies of `quizzes/summative.md` — one per student
- Rubric (`assessments/rubric.md`) for in-the-moment marking
- Timer

- Whiteboard

Lesson flow

Warm-up (3 min)

- Restate the safety rule one last time, chorally.
- Quick housekeeping: pairs sit together, decks in front of them, name and roll number on a slip clipped to the deck.

Summative quiz (20 min)

- Distribute the quiz papers. Pencils only. Closed book.
- Strict timing: 20 minutes.
- Collect at minute 20.

Project presentations (20 min)

- 2 minutes per pair (10 pairs ≈ full class; if more, adjust to 1:30 each).
- Each pair picks **one card** from their deck to present in detail. The other five are passed around or laid out for inspection.
- Mark rubrics in real-time. Don't try to mark after class — the energy fades.
- Q&A after each presentation is **30 seconds**: one classmate's question, one pair's answer. Move on.

Wrap-up (2 min)

- Collect decks for archiving (or return to pairs if they want them home — they earned them).
- One closing reflection — go around the room, each student in 5 words: *"What stays with you?"*
- Close: *"Three things to remember from this module: food amounts are safe; tradition and science can describe the same biological reality; ask 'how much' before you ask anything else about an herb."*

Student-facing content

Today is showcase + quiz.

1. Closed-book quiz, 20 min, 25 marks.
2. Present your favourite card from the deck for 2 minutes. Be ready for one quick question.
3. Leave with: the safety rule in your head, your deck in your hand, one fact you didn't know two weeks ago.

Differentiation

- **Tier up:** Volunteers can present **for** 3 minutes by request, presenting two cards — one of the six standard and one bonus card they made.
- **Tier down:** A student who freezes can have their pair-partner present while they hold up the card. Don't make this a failure mode.

Teacher notes

- The mark on the rubric **as the pair presents** — not after. Trust your first read.
- If the quiz takes longer than 20 minutes for the slowest students, extend by 5 max. Don't let it eat into presentations.
- Reflection question at the end ("what stays with you?") is the most honest indicator of what landed. Note it down if you can — useful for adjusting Year 2.

Citation(s) used in this lesson

- Quiz: `quizzes/summative.md`
- Rubric: `assessments/rubric.md`
- Project brief: `assessments/project-brief.md`

Writing Prompts

Writing Prompts — Herbs — Auṣadhi (Middle)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 200-word response.

1. In 3–4 sentences, explain Herbs — Auṣadhi to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Herbs — Auṣadhi with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Herbs — Auṣadhi? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Herbs — Auṣadhi (Middle)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — What is Auṣadhi?** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Tulsī — Holy Basil** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Neem — Bitter Healer** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Turmeric — Golden Root** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Ginger — Warming Root** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Ajwain — Carom Seed** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Mint — Cooling Leaf** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Herbs and the Doṣas** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Build the Class Deck** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.